CAFÉ FAUSSE

Web Application & Interface Design

Project Stats:

- Pages: 5
- API Endpoints: 8
- DB Tables: 2

- MSSE Project | React + Flask + PostgreSQL

PROJECT OVERVIEW

(I) THE CHALLENGE

- Café Fausse needed a full web presence
- Manual reservation calls overwhelmed staff
- No online customer data collection
- Required 5 complete pages with modern UI
- Needed backend database integration

PROJECT OVERVIEW

(II) OUR SOLUTION

- Full React SPA with 5 pages and routing
- Online reservation system
- Flask REST API with 8 endpoints
- PostgreSQL with Customers and Reservations tables
- Responsive design . mobile to desktop

TECHNOLOGY STACK

(I) FRONTEND

- React Router DOM
- Axios (HTTP Client)
- React Icons
- CSS Grid and Flexbox
- React 18 with Hooks

TECHNOLOGY STACK

(II) BACKEND

- Flask REST API
- SQLAlchemy ORM
- Flask-CORS
- Input Validation
- Error Handling

TECHNOLOGY STACK

(III) DATABASE

- PostgreSQL
- Customers Table
- Reservations Table
- Foreign Keys
- Auto Timestamps

WEBSITE PAGES & NAVIGATION

01) Home

Hero section, restaurant features, awards and positive customer reviews .

02) Menu

Full menu with category filters . appetizers, mains, desserts and drinks .

03) Reservations

Live reservation form with timeslot selection, guest count and table assignment**.**

04) About Us

Founder profiles, restaurant story, values and team highlights.

05) Gallery

Photo gallery with lightbox viewer showcasing the dining experience.

RESERVATION SYSTEM

Flow:

Customer Fills Form → React Validates → Flask API Processes → Table Assigned → Confirmation Sent .

(I) Form Fields Collected:

- Customer Name (required)
- Email Address (required)
- Phone Number (optional)
- Date and Timeslot (required)
- Number of Guests (required)
- Newsletter Opt-in

RESERVATION SYSTEM

(II) Backend Logic:

- Check timeslot availability
- Insert customer into Customers table
- Randomly assign an available table number
- Insert into Reservations with Foreign Key to Customer
- Return success or conflict response to React frontend

CUSTOMERS TABLE

Field	Type
customer_id	SERIAL (Primary Key)
ustomer_name	VARCHAR(255)
ustomer_email	VARCHAR(255)
phone_number	VARCHAR(20)
newsletter_signup	BOOLEAN
created_at	TIMESTAMP

RESERVATIONS TABLE

Field	Type
reservation_id	SERIAL (Primary Key)
customer_id	INT (Foreign Key → Customers)
time_slot	TIMESTAMP
table_number	INTEGER
guests	INTEGER
created_at	TIMESTAMP

FLASK REST API

Method	Endpoint	Description
GET	/api/health	Health check endpoint
GET	/api/menu	Retrieve full menu with categories
GET	/api/timeslots	Get available reservation timeslots
POST	/api/reservations	Create new reservation and assign table
GET	/api/reservations/<id>	Retrieve reservation by ID
POST	/api/newsletter	Newsletter email sign-up
GET	/api/customers	List all customers (admin)
DELETE	/api/reservations/<id>	Cancel a reservation

UI/UX DESIGN & KEY FEATURES

(I) Responsive Design

Fully mobile-optimized using CSS Grid and Flexbox. Tested across mobile, tablet, and desktop breakpoints.

(II) React Router SPA

Single Page App with client-side routing .no page reloads, fast navigation between all 5 pages.

(III) Form Validation

Client-side validation on all forms. Required field checks, email format validation, and error messaging.

UI/UX DESIGN & KEY FEATURES

(IV) Gallery Lightbox

Interactive photo gallery with lightbox viewer — clicking images opens a full-screen modal viewer.

(V) Newsletter Signup

Email collection form with backend storage. Validates email format and stores in PostgreSQL Customers table.

(VI) Awards & Reviews

Homepage showcases accolades and verified customer testimonials to build trust and credibility.